

# SPICESat: A Nanosatellite Mission to Explore the Propellant Sloshing Problem

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Many modern spacecraft and launch vehicles rely on large propellant tanks to perform attitude control and station-keeping maneuvers. When a large percentage of the spacecraft mass is comprised of a liquid propellant, sloshing becomes a critical aspect of spacecraft control and stability. SPICESat (Sloshing Platform for In-Orbit Controller Experimentation) aims to examine the issue of zero-g propellant sloshing from an experimental standpoint. The mission will study the tank fluid dynamics using new methods to gain an enhanced understanding of low-gravity fluid disturbance effects and improve equivalent mechanical models (EMMs). Active control of the fluid leading to the reduction of propellant slosh settling times will improve maneuverability and performance of spacecraft. In this paper, satellite mission objectives will be discussed, results of numerical simulations for the proposed control algorithms are shown, and a preliminary satellite design is presented.

## Nomenclature

|                |   |                                                                                           |
|----------------|---|-------------------------------------------------------------------------------------------|
| $B_o$          | = | Bond Number dimensionless value measuring the gravitational versus surface tension forces |
| $I_{sat}$      | = | Moment of inertia of satellite                                                            |
| $\theta$       | = | Satellite angular position                                                                |
| $\Omega$       | = | Satellite angular velocity                                                                |
| $\dot{\Omega}$ | = | Satellite angular acceleration                                                            |
| $\Gamma_{RW}$  | = | Commanded control torque                                                                  |
| $\Gamma_s$     | = | Sloshing torque                                                                           |
| $\Gamma_d$     | = | External disturbances (i.e. atmospheric drag in LOE)                                      |
| $NN$           | = | Neural Network                                                                            |
| $EMM$          | = | Equivalent mechanical models for approximating fluid sloshing in a tank                   |
| $CFD$          | = | Computational fluid dynamics                                                              |
| $SPICESat$     | = | Sloshing Platform for In-Orbit Controller Experimentation                                 |
| $ADCS$         | = | Attitude Determination and Control System                                                 |

## I. Introduction

Accurate attitude, positioning, and control are critical aspects of successful spacecraft operations. Satellite maneuvering accuracy is subject to disturbances generated by unspent propellants in the satellite's fuel tanks during orbital maneuvers or desired attitude changes. Fine guidance, such as needed for telescope pointing over long exposures, and coarse guidance, required when moving a satellite from one orbit to another, are all subject to disturbances generated by unspent propellants in the satellite's fuel tanks. The James Webb Space Telescope launched in December 2021, is a

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highly accurate astronomical instrument requiring position stability accuracy of 6 milliarcseconds (6/1,000ths of an arc second) or 1.67 millionths of a degree. A single arcsecond =  $\frac{1}{3600}$  of a degree. The telescope contains tanks filled with three different propellants. According to the James Webb Space Telescope User Documentation [1], the maneuver rate is determined in part by the need to keep pointing accuracy settling times balanced with the need to reach the new pointing as soon as possible. For large slews between 25 arcseconds and 3 arcminutes, the slew rate is slower than for shorter or longer slews, to avoid exciting slosh modes of the propellant in the tanks. The propellant slosh is projected to take more than 20 minutes to dampen in some cases. Another example of sloshing disturbances occurred in March 2007 SpaceX Falcon 1 second launch attempt [2], where a control failure at T+5 min, during second stage burn due to propellant slosh initiated by stage 1/2 contact at staging. Falcon 1 did not use slosh baffles in the second stage tanks, as simulations done prior to flight indicated slosh instability was a low risk. Given that in space there are no gust or buffet effects, the simulations did not take into account a perturbation, as occurred due to the hard slew maneuver after stage separation. Similarly, the effect of slosh are noted as a contributing factor in the NEAR burn anomaly in December 1998, where a rendezvous burn resulted in near loss of the spacecraft [3]. NEAR's two oxidizer tanks did not have internal propellant management device (baffles or diaphragms) for slosh mitigation. Subsequent slosh studies resulted in in-flight modifications to the spacecraft's control algorithms. The sloshing problem was visually captured during on January 29, 1964 during a Saturn 1 mission (SA-5) showing exactly how a fuel behaves in microgravity immediately after engine shutdown. Similar footage is also available from the SpaceX CRS-4 Launch internal fuel tank camera showing fuel sloshing under orbital acceleration and immediately after engine shutdown. The dynamics are clearly complex and non-linear.

Controlling fuel sloshing has been an engineering design concern since the dawn of the space age. In partially filled fuel tanks, large quantities of fuel can move inside the tanks under translational (thruster) and rotational (reaction wheels) accelerations. The fuel slosh develops from the dynamics of a free liquid surface inside of a fuel tank. As a spacecraft performs orbital maneuvers or attitude corrections, free moving propellant in a partially filled tank can excite low frequency oscillatory modes as the free liquid surface undergoes complex motions imparting forces and moments on the vehicle. As noted by Bourdelle [4], fuel mass as a percentage of total spacecraft mass can be considerable, and therefore mitigating sloshing effects through proper control algorithms becomes an important aspect of spacecraft design.

## II. Methodology

### A. SPICESat Experimental NanoSatellite

SPICESat (Sloshing Platform for In-Orbit Controller Experimentation) will be a 6U satellite form factor, with roughly one-third of the satellite consisting of a small, sealed tank filled with an appropriate propellant analogous liquid. The satellite is expected to be built throughout 2023 and certified for a launch opportunity in 2024. Sloshing will be excited through a series of rotational maneuvers across all 3 axes to generate tank / fluid disturbance torques. We intend to explore mid-range Bond numbers  $1 < B_o < 10$ . Bond number is a dimensionless number measuring the importance of gravitational forces compared to surface tension forces. Bond numbers of ten or less are generally considered to represent "reduced" gravity, so SPICESat mission results will apply to satellites in micro-gravity conditions undergoing slew maneuvers. The mission will characterize these sloshing effect by measuring standard attitude parameters and several new data collection techniques. Three methods of fluid characterization are to be explored: Motion sensing with accelerometers and gyroscopes, a pressure "sensor map" of fluid / tank interactions, and recorded video footage, enhanced by computer vision analysis.

SPICESat will have 3 main objectives:

- Demonstrate the feasibility of using pressure sensor arrays to map tank / fluid dynamics
- Demonstrate the feasibility of using computer vision to analyze complex sloshing dynamics
- Demonstrate the slosh disturbances mitigation strategies using novel active control algorithms

To accurately determine the spacecraft's attitude, the experiment will require high precision measurements. An on-board motion sensing suite (MSS) is expected to measure rotational motion and disturbances. This sensor array will build upon the 2005 SloshSat FLEVO experiment configuration [5]. We will use highly sensitive accelerometers ( $10^{-5} \text{ m/s}^2$ ) and rate gyroscopes (TBD  $^\circ/\text{s}$ ) to measure changes in attitude and fluid forces on the spacecraft caused by sloshing in the tank. Input control torques of a satellite,  $\Gamma_c \approx 10^{-4} \text{ N-m}$  typically result in sloshing torques of  $\Gamma_c \approx 10^{-5}$

N-m, translating into accelerations between 1 - 9  $\mu\text{g}$ , close to the detection threshold of a typical accelerometer.

A liquid sensing suite (LSS) will create a coarse "pressure map" of the actual forces that the propellant imparts on the tank wall. Ideally, lining the entire tank would form a continuous sensing surface, however, this precludes recording of video. Therefore, the sensor array will consist of 8 strips, placed strategically to cover each degree of freedom. Capacitive sensors are most ideal for this application.

A video sensing suite (VSS) will record video of the sloshing effects and transmit this information to the ground for further processing. Lightweight and low power cameras are being explored since the experiment only requires knowledge of the location of the fluid. Because the goal of the camera is to support the already-collected data, black and white footage will be used to decrease data volume. Modern computer vision techniques will be employed to visualize fluid / tank wall interactions, fluid oscillations, and the interactions between the fluid and tank wall.

The satellite is expected to conduct up to 252 different experiences to gather sloshing data for download to a ground station and associated analysis software. Ground software will be created to analyze the MSS data and reconstruct the state space vector for comparison to existing Computational Fluid Dynamics (CFD) models, tune the CFD parameters representing the fluid, and further validate existing existing EMMs. Satellite state space, pressure, and vision data will be combined and used to train a machine learning (ML) model to predict sloshing effects. Each experiment will be repeated 3 times to ensure accurate and reproducible data sets. Reaction wheel control inputs will be exercised along each axis of the satellite individually (x,y,z), then simultaneously along two axes (xy,xz,yz), and finally along all 3 axes at once (xyz), for a total of seven different excitation modes. Each experiment will measure a baseline control with no slosh mitigation strategies enabled, then a post excitation mitigation strategy to dampen the slosh effect, and then a combined excitation / mitigation active control strategy designed to reduce the effect of sloshing. Finally, the satellite will be designed to implement four different control algorithms. A baseline, standard ADCS controller will be used and will also act as the safe-mode controller for the remainder of the experiment. Next, a time delay control algorithm (also called an Output Feedback controller) will be implemented, where the controller uses a series of past values and implements a control strategy designed to mitigate sloshing. Then, a ground trained ML controller "black box" will be uploaded to the satellite and a control solution implemented using slosh predictions which will be used to reject the sloshing effects. Finally, the ML, time delay, and an add-on Reference Governor controller will be combined and tested. Reference Governors are used to ensure system dynamics respect a given set of constraints. Significant work has already been accomplished validating the EMMs [4], [6], and [7] and applying machine learning to satellite state vectors for predictions [8]. This ML approach and a description of the numerical results is highlighted in Section III.

## B. Experiment Equations of Motion and Flight Requirements

The basic equation of motion governing this problem are well known. Consider a spacecraft rotating rigid body in the absence of any disturbances:

$$I_{sat}\dot{\Omega} = \Gamma_{RW} \quad (1)$$

where  $I_{sat}$  is the moment of inertia of the satellite (single axis of rotation),  $\dot{\Omega}$  is the angular acceleration of the satellite, and  $\Gamma_{RW}$  is the torque generated by the reaction wheel. Taking sloshing and other disturbances (i.e. radiation pressure or LEO atmospheric drag) into account, the equation can be re-written as:

$$I_{sat}\dot{\Omega} = \Gamma_{RW} + \Gamma_s + \Gamma_d \quad (2)$$

where  $\Gamma_s$  is the sloshing torque and  $\Gamma_d$  represents any other external disturbances. Examining each term individually:

**Table 1 Satellite Sloshing Parameter Analysis**

| Parameter      | Measure  | Description                                          |
|----------------|----------|------------------------------------------------------|
| $I_{sat}$      | Known    | MoI matrix measured pre-flight                       |
| $\dot{\Omega}$ | Measured | Satellite angular acceleration measured in-flight    |
| $\Gamma_{RW}$  | Known    | Commanded control torque                             |
| $\Gamma_s$     | Unknown  | Sloshing torque                                      |
| $\Gamma_d$     | Unknown  | External disturbances (i.e. atmospheric drag in LOE) |

Since  $\Gamma_s$  and  $\Gamma_d$  cannot be easily distinguished from one another, this immediately places a number of requirements on SPICESat that must be considered. First, if we consider the entire system as a closed loop control system, we can generate a reaction wheel torque that compensates for both, such that:

$$I_{sat}\dot{\Omega} = \Gamma_{RW} + \Gamma_D \quad (3)$$

where

$$\Gamma_D = \Gamma_d + \Gamma_s \quad (4)$$

For the duration of each experiment, we assume:

$$\dot{\Gamma}_d = 0 \quad (5)$$

Although  $\Gamma_d$  is uncertain, measurements of atmospheric drag in LOE are expected to be small, between  $1 \times 10^{-8}$  km/s<sup>2</sup> and  $1 \times 10^{-10}$  km/s<sup>2</sup> and we are assuming them to be constant during the periods of our experiment, typically lasting 100 - 500 sec. Using the additional measurements methods proposed in this document, we aim to bound the interval of uncertainty in measuring  $\Gamma_s$ .

### 1. Experiment Considerations

In order to measure the effect of sloshing alone, the equations of motion (EoM) in Section B have placed key requirements on SPICESat that must be considered.

- **Altitude:** Launching to as high an altitude as possible ensures  $\Gamma_d \ll \Gamma_s$
  - **Sensor Sensitivity:** Employing extremely sensitive accelerometers and gyroscopes improves the ability to detect the effects of  $\Gamma_s$
  - **Alternative Measurements:** Must implement novel methods of detecting  $\Gamma_s$  such as video and pressure sensors
- Section II.D on Sensing Suites proposes three different methods for detecting  $\Gamma_s$

## C. Experiment Overview

SPICESat's primary experiment centers around a transparent, sealed tank containing a liquid analogous to spacecraft propellant, likely distilled water or silicone oil. To generate the necessary sloshing excitations with a Bond Number,  $1 < B_o < 10$ , the characteristic length scale of the tank is required to be sufficiently long. Therefore to extend the length of the tank, SPICESat must be a 6U satellite. The tank is expected to have 90 mm diameter and a length 160 mm of for a total volume 0.942 liters ( $0.000942 \text{ m}^3$ ) and a fill ratio of 70% or 0.709 liters ( $0.000709 \text{ m}^3$ ). Assuming a total 6U satellite mass of 7 kg and the fluid used is Dow Chemical silicone oil, the fluid mass represents just under 10% of the total SPICESat mass.

| Slosh Tank Physical Parameters |               |
|--------------------------------|---------------|
| Orientation                    | Horizontal    |
| Vessel Head                    | ASME F&D      |
| Shape                          | Torispherical |
| Diameter                       | 90.00 mm      |
| Vessel Head Depth              | 15.24 mm      |
| Straight Length                | 129.5 mm      |
| Total Tank Length              | 160 mm        |
| Dish Radius Parameter          | 1.00          |
| Knuckle Radius Parameter       | 0.06          |
| Head Type                      | Convex        |

## D. Sensing Suites

The satellite will be instrumented with three different measurement suites and a ground based sensor fusion algorithm. Each suite will be developed to facilitate the detection, data collection and transmission of the sloshing effects on the attitude of the spacecraft.

- MSS - Motion Sensing Suite
- VSS - Vision Sensing Suite
- LSS - Liquid Sensing Suite
- sFSS - Sensor Fusing Suite (ground)

### 1. Motion Sensing Suite (MSS)

The MSS will consist of a high precision accelerometers and gyroscopes (via the ADCS). The SLOSHSat FLEVO experiment flew in 2005 with six linear accelerometers with a sensitivity of  $1 \times 10^{-5} \text{ m/s}^2$ . SPICESat should consider this a minimum sensitivity and include 3 linear accelerometers and 3 low-drift gyroscopes on orthogonal axes oriented along the principle axes of the tank. Using slosh equivalent mechanical models and machine learning training conducted thus far, a minimum timestep,  $dt = 0.1 \text{ s}$  is required (10 Hz or better).

Assuming a typical input control torque of  $\Gamma_{RW} = 0.005 \text{ N-m}$ , the equivalent mechanical models indicate a sloshing torque of  $1 \times 10^{-4} < \Gamma_s < 5 \times 10^{-4} \text{ N-m}$ . Additionally assuming the ADCS is located 10 cm from the center of the tank (in a 6U satellite), then the order of magnitude of the force becomes:

$$F = \frac{\Gamma_{RW}}{L} = \frac{0.0001}{0.1} = 1 \times 10^{-3} \text{ N} \quad (6)$$

Again assuming a 6U satellite mass of 8 kg, this gives a minimum linear acceleration sensitivity requirement of:

$$a = \frac{F}{m} = \frac{0.001}{8} = 1.25 \times 10^{-4} \text{ m/s}^2 \quad (7)$$

**REQUIREMENT:** Linear accelerometer sensitivity =  $1.25 \times 10^{-4} \text{ m/s}^2 = 12.5 \mu\text{g}$

Validation work is underway to further the ADCS requirements and sensitivities using computational fluid dynamics (CFD) software, Flow-3D. Control inputs typical of Figure 3 are used to rotate the tank and excite sloshing. Center of Mass, fluid pressure, force, and velocity fields are calculated and will be used to ensure the components selected for the mission are suitable.

### 2. Vision Sensing Suite (VSS)

The VSS is a mid-resolution, modest frame-rate, black and white camera designed to capture the fluid motion inside the transparent tank. The choice of resolution and black and white video is driven by the desire to reduce data sent to the ground station. Additionally, color video adds very little information to the problem. Currently, several lightweight and small form-factor cameras are being evaluated, such as an Arducam.

Ground processing software is required to process the VSS video, using convolutional neural networks. The Burling lab has conducted very preliminary work using public domain software to detect and measure the surface disturbance of a sloshing liquid in 1g. This method can be used to measure and estimate the sloshing frequency needed as input to most EMMs. The software should also be able to detect interactions between the fluid and the side of the tank, as well as the approximate location. Tying this visual information with pressure sensor data and MSS measurements should permit sloshing identification.

To estimate the data storage requirements for the video, we use a reputable vendor calculator [9] with the following input parameters. These conservative estimates will need to be refined based on the camera selection:

| Video Camera Parameters |                   |
|-------------------------|-------------------|
| # Cameras               | 1                 |
| Resolution              | 960H x 540V       |
| Recording Time          | 600 s (0.138 hrs) |
| Frames/Sec              | 24 fps            |
| # Experiments           | 1                 |
| Compression             | H.264             |

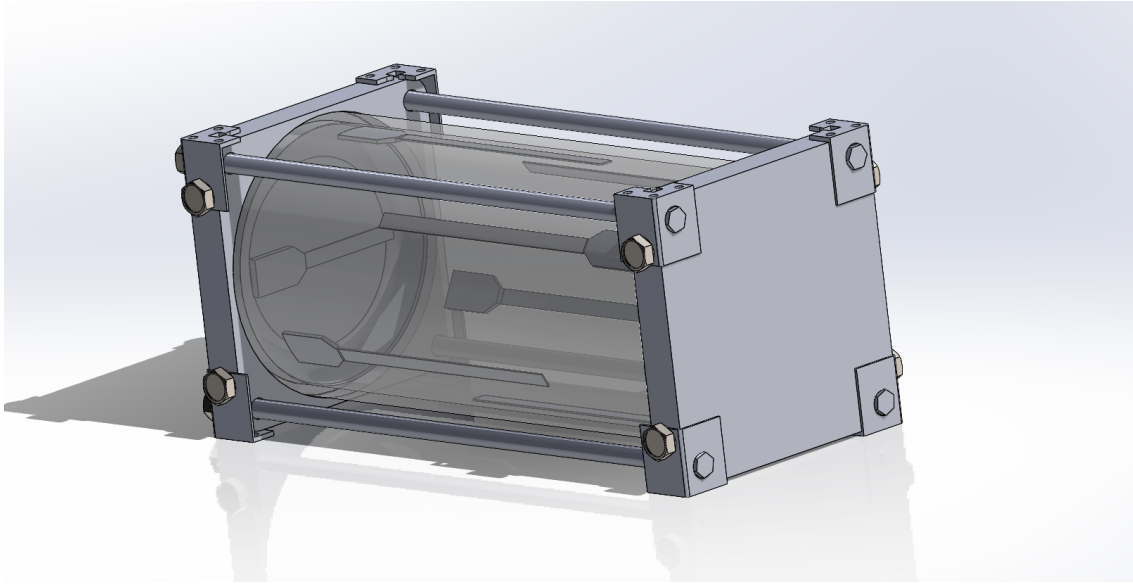
The resulting data file is close to 104 MB per experiment for image data. Accounting for communications overhead, this comes to 127MB per experiment.

### 3. Liquid Sensing Suite (LSS)

A second proposed method of detecting sloshing within a propellant tank is the use of pressure sensors to measure a quantity proportional to the sloshing torque. Then, using this measurement, an attitude controller can be designed not only to reject the sloshing torque but also to keep the fuel sloshing within acceptable conditions. Technically speaking, this is achieved by combining model-free control with an output to input saturation transformation (OIST) add-on. A first attempt to use pressure sensors in a tank is researched by Zhao, Fogel, and Burlion [10] using a sloshing equivalent mechanical model (EMM). The authors have shown that using a standard PID controller and adding a pressure measurement constraint to that controller, they can limit the sloshing effect to within a predetermined min/max.

The cubesat will be instrumented with pressure sensors located uniformly throughout the tank. Figure 1 shows a schematic illustration of the proposed sensors inside the tank (left) and tank within the cubesat (right).

**Fig. 1 Illustration of pressure sensor array inside tank**



A pressure sensor array of at least eight sensors will line the inside of the tank. Inserted horizontally, the sensors will divide the tank into sectors with sensors at the ends of the tank. This configuration allows for vertical and horizontal measurements of the tank-fluid interaction (Figure 1). Capacitive based pressure sensors, specifically the model 5420 (or equivalent) from Pressure Profile Systems (PPS) Inc. is a strong candidate. Each sensor is waterproof and has a low power consumption. From preliminary analysis, sloshing motion can be captured at 10 Hz. Each sensor array will output  $\sim 0.3$  kB/s of data at a 10 Hz scan rate. The PPS sensors are ideal for this application because of their ability to capture the full complexity of the dynamics of the system. This produces a coarse pressure map, captured in real-time, of the interaction between the fuel and the tank wall. The PPS sensors will be connected in a “daisy chain” configuration with associated conditioning electronics which output standard SPI protocol, which will communicate with the on-board computer. The flight computer will then handle any additional data processing and storage required. Sensor data will be used for active control feedback as outlined in Section II.E.

Sensor testing will be conducted in house with known slosh models to verify that accurate measurements are maintained. Additional experimental testing and verification will take place using the Rutgers slosh test bench experiment. A zero-G flight will test flight conditions for the fully integrated sloshing payload in 2023. The results of this test will affirm final checks for mission readiness.

### 4. Total Data Requirements

During the lifetime of the experiment, we expect to conduct up to 252 experiments, including 3 trials of each experiment, 7 different rotations, 3 satellite modes across 4 different controller types. One-third of those experiments (84) will include camera data and the remaining 168 experiments will include sensor data only. The MSS, VSS, and LSS all contribute to the total experiment data download requirement.

**REQUIREMENT with Camera:** Total Data Per Experiment = MSS(28.84 Mbits) + VSS (1016.00 Mbits) + LSS (115.2 Mbits) =

1160 Mbits / experiment = 145 MBytes / experiment (CAMERA)

**REQUIREMENT without Camera:** Total Data Per Experiment = MSS(27.84 Mbits) + LSS (115.2 Mbits) =

143.04 Mbits / experiment = 17.88 MBytes / experiment (NO CAMERA)

**84 Experiments @ 145 MB = 12,180 MB**

**168 Experiments @ 17.88 MB = 3004 MB**

**ALL experiments: 252 Experiments = 13,168 MB**

The satellite itself also transmits important State of Health (SoH) data to the ground in addition to the Experiment Data. This includes information such as battery charge over time, satellite temperature, ADCS statistics, etc. This SoH data is designed to ensure the non-experiment related components are functioning properly. Therefore, in total, the data download requirements include both SoH and Experiment information.

**REQUIREMENT with SoH & Camera:** Total Data = Experiment(145 MBytes) + SoH (7 MBytes) =

**152 MBytes / experiment TOTAL**

### 5. Sensor Fusion Sensing Suite (sFSS)

Finally, all sensor data is expected to be recorded and transmitted to the ground and analyzed. All data should be time-stamped and synchronized on-board SPICESat with the intention of comparing satellite motion (MSS), the pressure measurements (LSS) by location within the tank, and the corresponding visual interaction (VSS) of the fluid with the side of the tank. By fusing all the data into a single analysis, we intend recover full satellite state space information including:

- $\theta$  – Angular position
- $\dot{\Omega}$  – Angular velocity
- $\ddot{\Omega}$  – Angular acceleration
- $\Gamma_s + \Gamma_d$  – Sloshing torque disturbance + external torque disturbance
- $\dot{\Gamma}_s$  – Sloshing torque disturbance rate of change
- $\Gamma_{RW}$  – Known input control torque (reaction wheel)

Once the state vector is recovered from the sensors, the data can be used to validate the existing EMMs and used to re-design and upload new control strategies to SPICESat.

## E. Experiment Operating Modes and Design

### 1. Slosh Excitation

The scientific portion of this project requires SPICESat to maneuver in various directions and attitudes to generate sloshing modes inside the cylindrical tank. Critical to understanding sloshing characteristics are maneuvers that mimic larger satellite flight characteristics. This will require a set of pre-programmed maneuvers for SPICESat and similar in nature to the SloshSat experiment [5]. The satellite will cycle through each of the maneuvers for every new controller algorithm tested. Each experiment should be started from an initial condition rest state of zero fluid velocity within the tank and conclude when the fluid is at rest again. Subsequent experiments should only begin after the fluid is completely settled.

- **Rotation About a Single Axis** – Spin the tank at a specific angular rotation rate along on single axis (x,y,z)
- **Rotation about Two Axes** – Spin the tank about two axes (xy, yz, xz)
- **Nutation About Multiple Axes** – Spin the satellite about multiple axes at the same time to simulate tumble recover (xyz)

To ensure adequate control authority throughout each experiment, we have conducted both EMM calculations and are currently performing CFD calculations using Flow-3D to determine the total forces and torques the sloshing fluid will exert on the satellite. The EMMs indicate a sloshing torque on the order of  $0.1mNm$ . The currently selected ADCS has a maximum control torque of  $5mNm$ . Early work using Flow-3D validate this approach, showing a sloshing torque of  $0.9mNm$ .

## 2. Experiment Operating Modes

The experiment will require multiple satellite operating modes to accommodate science objectives.

- **Excitation** – Excite sloshing using reaction wheels
- **Observation** – Enable all sensor suites, actively observe, and record results
- **Mitigation** – Employ control algorithms to actively dampen sloshing and reduce settling times
- **Excitation & Mitigation** – Controlled excitation of sloshing while simultaneously using active control algorithms to mitigate slosh
- **Idle** – All maneuvers and observations complete, fluid settles until next experiment
- **Recovery** – Fallback safe mode controller to bring the satellite and fluid back under control in the event of excess excitation
- **Algorithm Update** – Enable the attitude control system to accept a newly uploaded control algorithm

## F. Slosh Mitigation Algorithms

After a period of data gathering and sensor analysis on the ground using system identification techniques like SINDy (Sparse Identification of Nonlinear Dynamics) and DMD (Dynamic Mode Decomposition), various control algorithms will be explored to mitigate slosh effects on the satellite. A primary output of this data collection and analysis is to iterate through various control algorithms using ground simulations. This goal is to maximize slosh mitigation performance. On the ground, we will have two methods to test these controllers, 1) Numerical simulations using equivalent mechanical models and, 2) a 1-g, 1 DoF test bench experiment. Some control algorithms will also employ "CFD in the loop" testing prior to upload to the satellite.

As an example, the data used from the satellite can be directly used to train a machine learning (ML) algorithm. Using the MSS data and the known initial control input, a ML model can be trained to predict the sloshing effect, and therefore compensate for it in the control feedback loop. This is discussed in more detail in Section III.B.

This iterative approach to satellite control requires the flight control software have the capability to upload new algorithms while always preserving a standard controller in case of recovery situation.

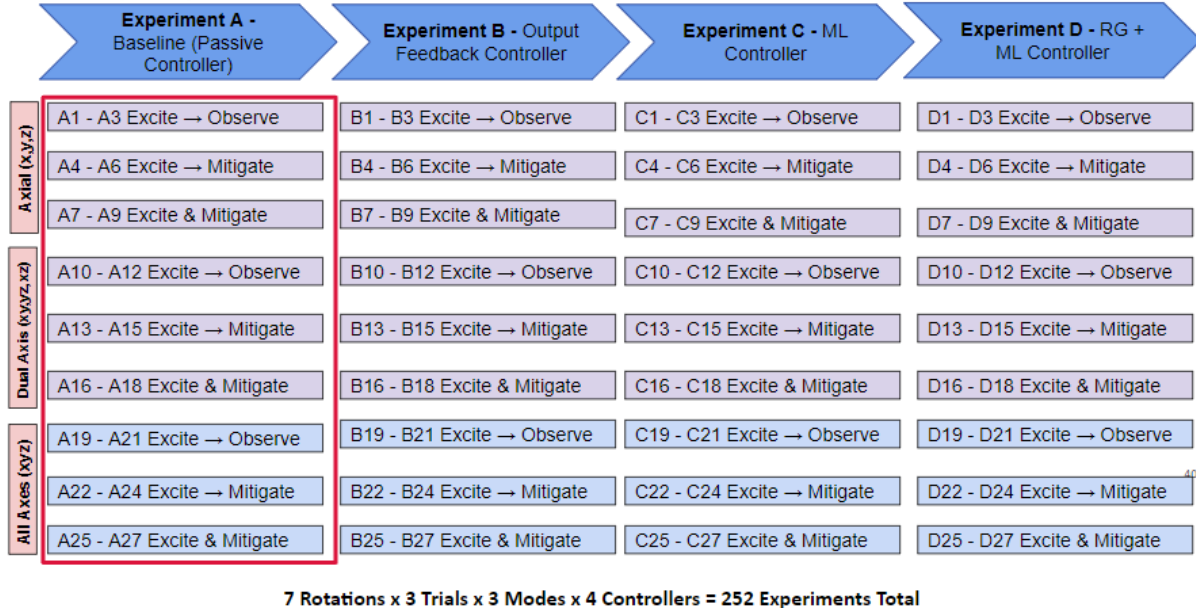
- 1) **Standard Controller** – Known, traditional, safe PID controller for satellite recovery
- 2) **Output Feedback Adaptive Control** – Uses past sensor and state vector values to modify an existing PID controller based on real-time output from the satellite MSS
- 3) **Machine Learning Controller** – Using the recorded LSS, VSS, and MSS, pre-train a Neural Network algorithms to predict sloshing effects for a given control input
- 4) **Reference Governor Controller** – A type of controller that is added onto any pre-existing controller and ensures control inputs and measured outputs remain within safe operational constraints.
- 5) **Machine Learning + Adaptive + Reference Governor Controller** – Fuse the key controller algorithms into a single adaptive controller

## G. Experiment Phases

The experiment requirements outlined in Section II.D.4 will be carried out across four different controllers, 3 different modes, 7 different excitations, and repeated 3 times. Therefore there will be a total of 252 experiments performed throughout the flight. Experiment Phase A will represent the minimum success criteria for the entire mission.



**Fig. 2 Schematic Diagram Representing Experiment Plan. Red box represents minimum mission success criteria**



Each phase of the experiment, A thru D, will have the following items specified:

- **Objectives** - Different for each phase based on each controller type being tested
- **Entry Constraints** - Very similar across all experiments - No slosh, ADCS ready, accel / gyro steady, liquid sensors ready, camera ready; different excitation preparation - axial rotation, point rotation, nutation; controller health is verified
- **Success Criteria** - Successful excitation and recording of sensor measurements
- **Data Products** - All data from sensing suites listed in Section II.D
- **Ground Config** - Communications ready for satellite passing overhead and sFSS ready to accept and process data
- **Spacecraft Config** - Changes based on excitation requirement

### III. Numerical Results

#### A. Equivalent Mechanical Models

Attitude changes in a satellite are typically accomplished by the use of reaction control wheels or magnetorquers. As a result, we can assume that the mass and therefore the fill ratio of the propellant remains constant. The satellites under consideration are assumed to be in microgravity, so we can neglect the effects of gravity. We also assume that there are no external perturbations, so that  $\Gamma_d = 0$ . The tank is rigid and in a fixed position. As the satellite changes attitude, the angular velocity and acceleration imparted by the control wheels is the predominant influence on the propellant. A final assumption made in this research is neglecting noisy data from the measured states. Real world measurements of angular velocity and acceleration and associated signal to noise ratios should be considered in future research.

Following the treatment by Bourdelle [4] [7], we can define an equivalent mechanical model (EMM) to represent the fuel sloshing behavior using a second order non-linear differential equation:

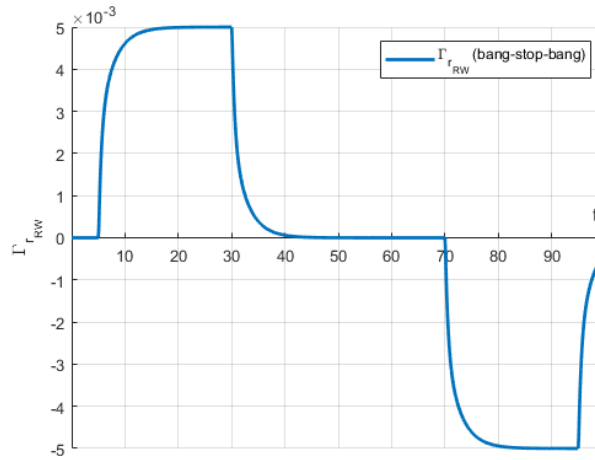
$$\ddot{\Gamma}_s(t) = -A_s(t)\Omega(t) - B_s(t)\dot{\Omega}(t) - C_s(t)\dot{\Gamma}_s(t) - K_s(t)\Gamma_s(t) \quad (8)$$

$$\dot{\Omega}(t) = \frac{1}{I_{sat}}(\Gamma_{RW}(t) + \Gamma_s(t)) \quad (9)$$

Input control torques are typically a "bang-stop-bang" profile, where a reaction wheel is spun up and then stopped, imparting its momentum back into the body of the satellite and forcing it to rotate on one axis. Then the reaction

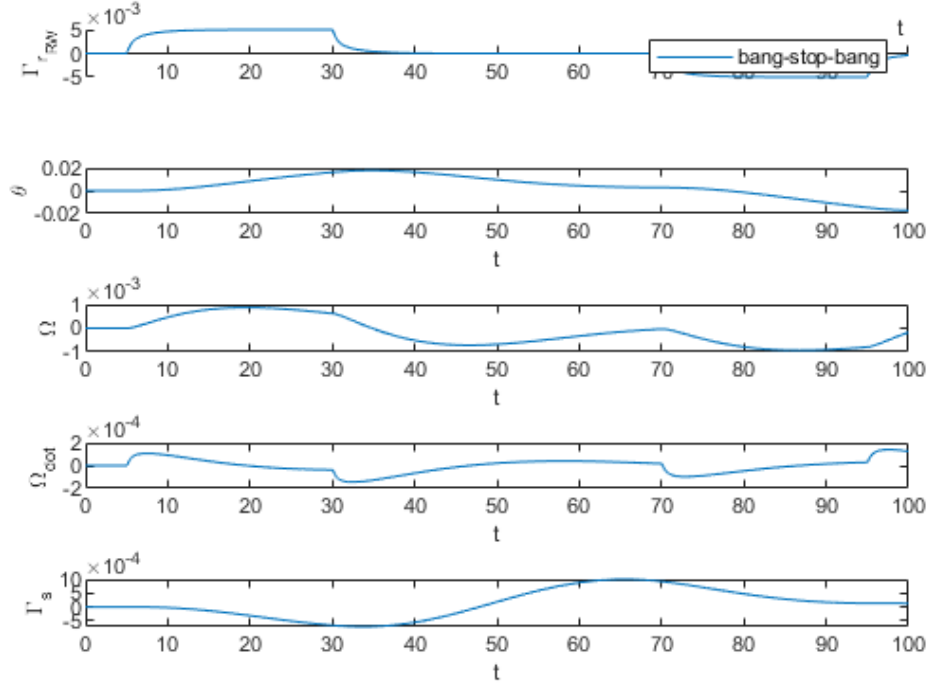
wheel is spun up again in the opposite direction until the satellite reaches the desired change of angle and stopped again, imparting a torque in the opposite direction this time, thereby stopping the satellite rotation. This is why such a maneuver is called a "bang-stop-bang" profile. A typical order of magnitude for  $\Gamma_{RW} < 0.005$  Nm. In practice, these bang-stop-bang profiles are generated by reaction wheels that have a characteristic response curve. We have approximated how these low-pass dynamics are modeled by the following transfer function:

$$H(s) = \frac{1.2s + 0.76}{s^2 + 2.4s + 0.76} \quad (10)$$



**Fig. 3 Typical reference control torque with actuator transfer function added. ( $\Gamma_{RW}$ ) "bang-stop-bang" profile.**

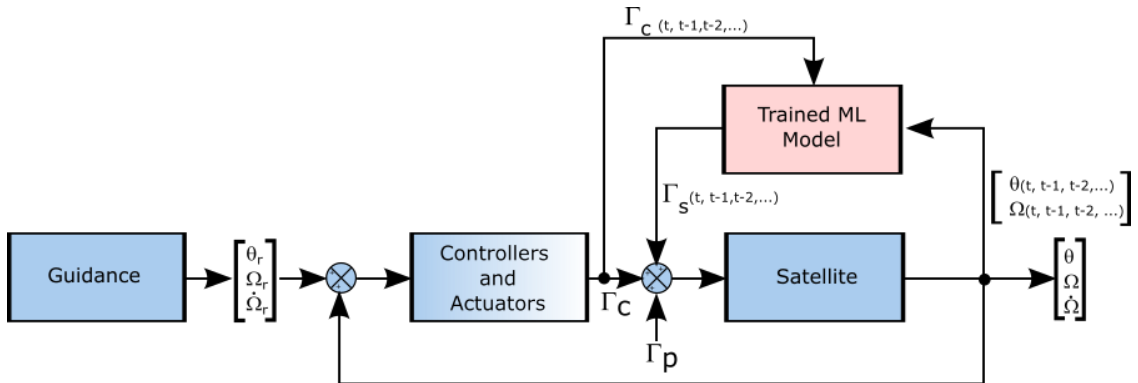
The dynamics from Equations 8 and 9 are modeled in Simulink where we solve for the sloshing torque,  $\Gamma_s$ . Figure 4 shows the results for a typical input torque profile.



**Fig. 4** Simulink output showing attitude ( $\theta$ ), angular velocity ( $\Omega$ ), angular acceleration ( $\dot{\Omega}$ ), and sloshing torque ( $\Gamma_s$ ) for a the typical control torque ( $\Gamma_{RW}$ ) "bang-stop-bang" profile

### B. Machine Learning for Control

In [8], we have examined the feasibility of predicting sloshing behavior on-board satellites during maneuvers using machine learning and using these predictions to design a control algorithm to mitigate the disturbance effect. We do this by training a ML algorithm on a set of input control torques,  $\Gamma_{RW}$  and state vector responses. Once trained, we can exchanging the EMM in our plant with a prediction for the sloshing torque  $\Gamma_s$ :



**Fig. 5** Control block diagram using a ML block for predicting  $\Gamma_s$ .

In this case, the dynamics of the system are completely replaced by a trained "Black Box" knowing nothing about the physics of the problem. The ML model is previously trained offline from the control algorithm. This now smart "Black Box" takes inputs from the system ( $\theta(t)$ ,  $\Omega(t)$ ,  $\Gamma_c(t)$ ) and predicts the effect of sloshing on the system, outputting

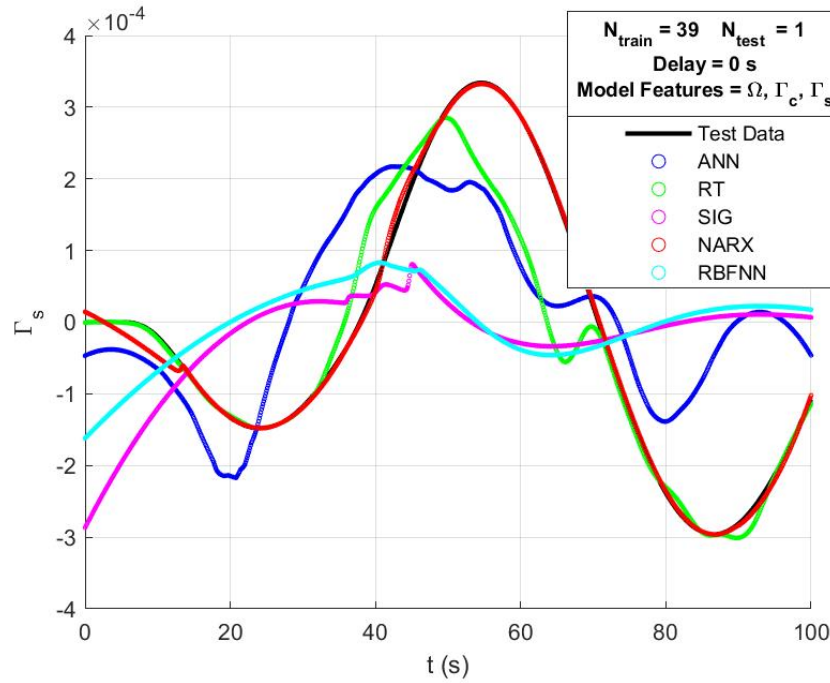
$\Gamma_s(t)$ . This sloshing prediction is then added to the existing PD controller and a new control torque  $\Gamma_c$  is calculated.

### C. Machine Learning and Prediction Results

In the broad and growing field of ML, there are a large number of ML algorithms that can be applied.

- Sigmoid (SIG)
- Radial Basis Function Neural Networks (RBFNN)
- Regression Trees (RT)
- Artificial Neural Networks (ANN)
- Nonlinear Autoregressive Exogenous Model (NARX)

After the Model Features are loaded into the associated data sets and the training algorithms designed, the time consuming process of training the models against the data begins. Figure 6 shows a typical result of fitting the 5 different algorithms to two of the measured states,  $[\Omega, \Gamma_c]$  and attempting to predict the slosh Response,  $\Gamma_s$ . The test Response is the black line and is the goal of all the NN models to try to find the best fit for.



**Fig. 6 Neural Network fit - No Delay**

These numerical results show that NARX neural networks (Nonlinear Autoregressive Exogenous Models) are best suited for application to this problem. This make sense, because NARX networks are commonly used in time-series modeling. For SPICESat, we propose to use the actual excitation data recorded onboard the satellite and use both Neural Network and System Identification techniques to train and predict sloshing response, which we will call  $\Gamma_{sNN}(t)$  to reject this disturbance using a control law:

$$\Gamma_{RW}(t) = -K_p\theta(t) - K_d\Omega(t) - \Gamma_{sNN}(t) + r(t) \quad (11)$$

## IV. Satellite Design and the University Nanosatellite Program

### A. The University Nanosatellite Program

In October of 2021, the Air Force Research Lab in Albuquerque, New Mexico selected Rutgers University and the SPICESat mission for Phase A funding through its University Nanosatellite Program (UNP). The UNP funds undergraduate university students to design, build, launch, and operate small satellites. Phase A lasts for two years, ending with a down-selection at the Flight Selection Review (FSR). During those two years, the SPICESat team is held to tight schedule, similar in nature to a commercial or government satellite design program.

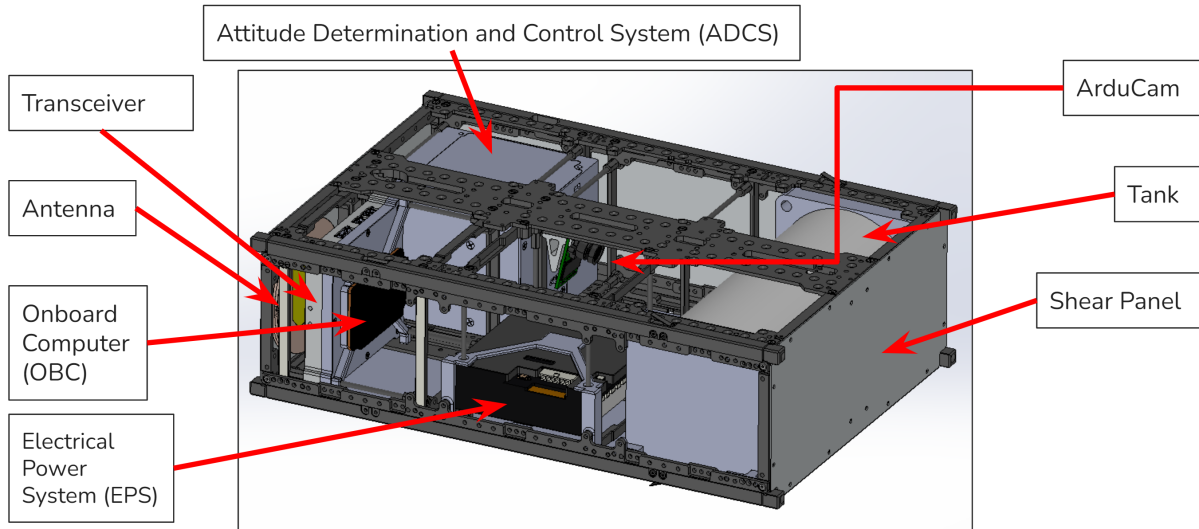
**Table 2 SPICESat Design Review Process**

| Phase    | Description                  | Time Frame               |
|----------|------------------------------|--------------------------|
| Kick-off | Program Kick-off Meeting     | January 2022             |
| SCR      | System Concept Review        | February 2022            |
| SRR      | System Requirements Review   | April 2022               |
| PMR      | Program Management Review    | August 2022              |
| PDR      | Preliminary Design Review    | November 29, 2022        |
| FAB      | Satellite Fabrication Course | August 2023              |
| CDR      | Critical Design Review       | September / October 2023 |
| FSR      | Flight Selection Review      | January 2024             |

### B. SPICESat Preliminary Design

SPICESat recently conducted its Preliminary Design Review (PDR). That review included a number of initial design review concepts focusing on the individual components required to fulfill the mission. Figure 7 shows the most recent satellite configuration with all key components included. Solar panels are removed in order to show a more complete internal diagram.

**Fig. 7 Satellite Preliminary Design Concept**



Each component has been analyzed and compared against the mission requirements as detailed in Section II. Many of the components listed below have been ordered and satellite engineering model construction should begin in January 2023 with a "flat sat" satellite expected to be operational by August 2023.

**Table 3 SPICESat Component Overview**

| Component              | Description                    | Vendor                        |
|------------------------|--------------------------------|-------------------------------|
| Control (MSS)          | Satellite attitude and control | Blue Canyon XACT-50           |
| Camera (VSS)           | Sloshing video recording       | ArduCam OV2640                |
| Tank                   | Primary experiment with fluid  | ALON from Surmet              |
| Pressure Sensors (LSS) | Pressure sensor mapping        | Pressure Profile Systems 5420 |
| Chassis-Panels         | Preliminary Design Review      | SpaceMind                     |
| Electrical-Power       | Power for satellite            | SpaceMind EPS                 |
| Onboard Computer       | Main computer                  | GOMSpace NanoMind A3200       |
| Antenna-Receiver       | Ground communications          | S-band Transceiver            |

## V. Conclusions and Future Work

SPICESat promises to be a very complicated and interesting mission with the capability to validate existing models and provide a new method for attitude control in space. Previous experiments in this area lack detailed understanding of internal tank dynamics, which we propose to detect using internal pressure sensors and camera footage. This collected data will be able to be used to validate equivalent mechanical models and create new, more representative ones. With the new insight, new control algorithms will be trained using classical methods or machine learning before being updated on-board.

In the case of imaging and observatory military satellites with large propellant tanks, this research translates to faster slew rates, more agile maneuvers, and ultimately, higher performance. Beside observation spacecraft, development and implementation of new control methods using Machine Learning to solve the “hard” problem of fuel sloshing will directly benefit future DoD aerospace platforms subjected to various kind of disturbances and modeling uncertainties.

Future work includes the initial construction of a “flat sat” where each component will be laid out on a flat board test bench and wired as required. Then software and hardware will be tested to ensure compatibility and suitability against the mission requirements.

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